

2026 Net Zero Tech — Summary Table of Project Descriptions

Track: Sustainability — (1) Carbon Management & Climate Tech (Carbon Removal / Negative Emissions, MRV). Format: Arial 12, 18 pt line spacing, A4. Fields in 【 】 are placeholders.

No.	(assigned by organizer)
Work Title	LeafCarbon: low-cost, non-destructive estimation of plant carbon from images
School	National Taiwan University (leader & CS members); National Taipei University of Technology (materials members)
Department	Computer Science and Information Engineering (NTU); Materials (National Taipei University of Technology)
Advisor	Prof. Dong-Hau Kuo — National Taiwan University of Science and Technology, Dept. of Materials Science and Engineering; Tel +886-2-2730-3291; dhkuo@mail.ntust.edu.tw
Leader	CHEN, JING-SHENG (NTU, CSIE); Tel 0968739213; d14922013@ntu.edu.tw
Member	CHUANG, ETHAN LIU (NTU, CSIE), 0966431050, r13922a24@csie.ntu.edu.tw; ZHANG, ZHONG-WEI (Taipei Tech, Materials), 【phone/e-mail TBD】; SU, YOU-QI (Taipei Tech, Materials), 【phone/e-mail TBD】

1. Introduction of Team

We are a Computer Science and Materials Science student team. CS works on the image machine-learning model, the confidence-based routing logic, and the demo system. Materials works on the oven dry-weight calibration, the imaging standardization, and the measurement procedure. A measurement needs a model and reference values to check it against; CS does the first, Materials the second.

2. Creation Motive — *problem-solving and technical value*

For biological carbon removal (afforestation, soil, restoration, microalgae), the hard part is the measurement: knowing how much carbon was captured, accurately and cheaply. This is the MRV step (Monitoring, Reporting, Verification). The standard route is destructive harvest, oven-drying, weighing, then lab analysis, sample by sample, which is too expensive for smallholders or large areas. So operators who already capture carbon often cannot earn credits, because the amount is hard to measure affordably and its permanence hard to prove. Taiwan's carbon fee (NT\$300/t CO₂e from 2025) and its 2040 target of +10 Mt CO₂e of farm sinks both depend on this measurement. Its official forest-carbon methodology (Forestry Research Institute, 2025) sets the bar (third-party verification, uncertainty below 10%) but covers forests only. Soil and biochar already have international methodologies (Verra VM0042 and VM0044); what they lack is affordable, fine-grained measurement, because these rely on lab soil sampling or coarse satellite remote sensing. LeafCarbon is a low-cost way to fill that measurement gap.

3. Research Process

(1) We built our own dataset: lettuce we grew ourselves, photographed day by day, paired with dry weights and on-site environment records. (2) We segmented the plant out of the low-cost images (green-canopy extraction) and computed structure features (canopy area, plant height, canopy density). (3) We added environmental history (cumulative light, growing-degree-days). (4) We regressed per-plant dry biomass with a Random Forest and validated it with leave-one-day-out cross-validation. (5) We anchored the model to ground truth with a small set of oven-dried dry weights. Carbon is then taken as dry weight $\times 0.47$, a widely used constant, so elemental analysis is not required. Each estimate carries a confidence interval, so only uncertain samples are sent to the lab.

4. Brief of Work — *creativity / technical content and feasibility*

LeafCarbon is a measurement setup: standardized low-cost imaging, an AI estimator, and a small calibration set. The carbon fraction of plant dry matter is close to constant (about 47%; IPCC 2006 default 0.47 tC/t; Ma et al. 2018 report 45.0–47.9% across 4,318 species). The quantity that actually varies is biomass, so carbon = biomass $\times 0.47$. The image estimates biomass, and the small set of oven-dried weights calibrates the image-to-biomass relationship; carbon then follows from the 0.47 constant, with no elemental analysis. Estimating biomass from leaf structure is an established approach (for example NDVI and SPAD), so the method builds on validated work and our effort goes into the integration. What we add is a low-cost, non-destructive measurement tool set up for carbon projects. Lettuce is a fast validation crop here: its roughly 35-day cycle lets us tune the image-to-biomass relationship and validate the workflow quickly, and the same method is intended to transfer to durable pools such as forests, soil, and biochar. Unlike satellite digital-MRV tools such as Boomitra, which measure soil carbon over large areas and issue credits, LeafCarbon is a ground-level, single-plant, non-destructive measurement module that issues no credits and can sit upstream of any capture project.

5. Expected Benefit — *specific expected efficiency, output value, data*

On our own lettuce image set (569 photos over days 7–35), the per-plant biomass model is trained and validated on the 72 single-plant samples that have individual dry weights. Using side-view structure features together with cumulative environment, it reaches $R^2 \approx 0.64$ and $RMSE \approx 0.43$ g under leave-one-day-out cross-validation. This is a measured result on our data, with the model fully specified (image and environment $\rightarrow$ biomass $\rightarrow$ carbon). On cost and speed: the traditional way to get one plant's biomass and carbon is destructive harvest, oven-drying, weighing, then lab elemental analysis (roughly US\$20–75 per sample), done sample by sample and never repeatable; LeafCarbon estimates it from a photo in seconds, without destroying the plant, and the same plant can be measured again later. At project scale, digital MRV reportedly cuts per-farm verification cost by large margins (Boomitra reports over 90% versus traditional sampling). This would let afforestation, soil-carbon, and restoration projects, and smallholders, afford verification, and it aligns with Taiwan's carbon fee and the 2040 agricultural-sink target.

6. Project Introduction Video (≤ 5 min, MPEG-4, English, download enabled)

Cloud link: <https://drive.google.com/file/d/1thV8Gi64WbX1wUuungYQzmBUGm3e4QiL/view?usp=sharing> (sharing set to "anyone with the link", download enabled). Flow: the problem (measuring carbon is too expensive to scale) → what we built (a low-cost imaging measurement setup) → footage (the capture rig and the oven dry-weight calibration) → AI demo (photo and environment → biomass and carbon, with a confidence interval) → measured results and a cost comparison.